Keyhole Labs releases open source Byzantine Tools for blockchain

The team at Keyhole Labs, the product and innovation arm of Keyhole Software, has announced the release of Byzantine Tools, a series of blockchain open source projects to enhance Hyperledger blockchain networks.

Keyhole Labs is a group within Keyhole dedicated to creating open source solutions that help software developers in their craft. Byzantine Tools is an extension of the company's open source initiatives with a more specific focus on blockchain technology.

The new release includes the Byzantine Browser, Byzantine Config and Byzantine Flu open source tools. All add functionality to Hyperledger blockchains or show examples demonstrating the importance of blockchain to enterprise-level organisations.

Byzantine Browser is an open source analytics tool for real-time visibility into transactions and blocks as they are added to a Hyperledger Fabric network.

Byzantine Config provides features related to configuration. Updating a Hyperledger Fabric network configuration manually with CLI tools can be complex and cumbersome. This open source utility provides an executable GUI application that simplifies the configuration process by producing a config transaction buffer file that can be signed and executed to update a Hyperledger blockchain configuration.

Byzantine Flu is a Hyperledger Fabric reference application deployed to the cloud that tracks the spread of influenza across a country.

According to the Keyhole Labs team, the features provided by blockchain technology can lead to business benefits like lower costs, higher efficiency, and lower risk through decentralisation and immutability.

"While cryptocurrencies brought blockchain to the forefront of the technology headlines, the technology underneath has true potential value for the enterprise, outside of the cryptocurrency space," the team members said.

businesses and organisations. We pledge to guard and maintain the Open Source Definition, and we recognise the Open Source Initiative as the steward of the Open Source Definition."

Five leading global organisations join the OpenSDS project

The Linux Foundation has announced that Click2Cloud, GMO Pepabo, Intel, the Internet Initiative of Japan, and KPN have joined the OpenSDS project.

The new contributors to the OpenSDS project will help accelerate its evolution, including the development of Flash management, and the adoption of NVMe and NVMe over Fabric in cloud native environments.

The OpenSDS community also announced the release of Bali, the second version of its open source software for software-defined storage control.

Bali features support for multi-cloud data control with a single S3 object REST API across multiple clouds, allowing end users to combine object storage services from AWS, Azure, and Huawei into a single solution.

The on-premise Ceph S3 object is in development and will be available as an update to the Bali release, the community said in a press release.

Bali also supports manual data migration from on-premise storage to cloud storage. Policy-based multi-cloud data control and support for Google, IBM, and other cloud service providers are planned for 2019.

Click2Cloud, a cloud migration technology company with expertise in cloud comparison, CMP, infrastructure automation and cloud migration, is a multinational corporation incorporated in the state of Washington.

GMO Pepabo Inc. is a Tokyo and Fukuoka-based technology company which focuses on consumer-targeted Internet services, whereas KPN is the leading supplier of telecom and IT services in the Netherlands.

IIJ (Internet Initiative of Japan), Japan's first ISP, believes that the collaboration with OpenSDS will help to accelerate the modernisation of its data centre.

Meanwhile, Intel is looking forward to working with the OpenSDS community to drive the project's evolution and adoption.

"We are excited to welcome so many leading global organisations into the OpenSDS project to help deliver on our vision of a vendor-neutral intelligent data storage management framework for the cloud era," said Steven Tan, OpenSDS TSC chairman, VP and CTO of cloud storage solutions at Huawei.

"Bali multi-cloud is another step forward in our path to offer end users the freedom to choose services that best suit their needs," Tan added.

Rakesh Jain, OpenSDS TSC vice-chair and senior researcher and architect at IBM, said that the objective for the next releases is to further place OpenSDS in the modern storage hierarchy, in particular with NVMEoF.

OpenSDS is the world's first open source community focused on software-defined storage. It is dedicated to providing a unified SDS controller framework and APIs for cross-cloud workloads. The community